

Jixian Li, Ph.D.

Research Associate, Texas Advanced Computing Center, University of Texas at Austin

✉ jixianli@tacc.utexas.edu ☎ +1(520)903-4340

🌐 <https://jixianli.github.io/>

🐙 GitHub 🔗 LinkedIn

Biography

As a research associate in the Scalable Computational Intelligence team at TACC, I collaborate with scientists to design and build intelligent systems that improve and automate scientific data analysis workflows. My goal is to make scientific discovery simple and efficient.

Previously, as a postdoctoral research associate at the Scientific Computing and Imaging (SCI) Institute under Dr. Chris Johnson, my work focused on developing innovative visualization methods for scientific data, including computational fluid dynamics, medical imaging, and biomedical simulations.

I received my Ph.D. from the University of Arizona, advised by Dr. Joshua Levine, where my dissertation studied deep learning approaches for exploring collections of visual features of scalar fields.

My research interests include visualization, uncertainty analysis, machine learning, and computational topology related to scientific data analysis. My broader goal is to help scientists to compute, analyze, and communicate their findings.

Education

2022	▶ Ph.D. Computer Science	University of Arizona
2017	▶ M.S. Computer Science	University of Arizona
2015	▶ B.S. Computer Science & Mathematics	University of Arizona

Employment History

2026-...	▶ Research Associate	Texas Advanced Computing Center
2022-2026	▶ Postdoctoral Research Associate	SCI Institute, University of Utah
2019	▶ Summer Intern	Lawrence Livermore National Laboratory
2017-2022	▶ Research Assistant	Dept. Computer Science, University of Arizona
2015-2017	▶ Teaching Assistant	Dept. Computer Science, University of Arizona

Publications

Forthcoming

- 1 K. Borkiewicz, J. Li, J. A. Levine, and K. E. Isaacs, "May (A)I beautify your visualization? expert judgments of acceptable aesthetic alterations," in *2026 IEEE Visualization and Visual Analytics (VIS)*, 2026.
- 2 T. Finken, J. Li, B. Wang, H. Guo, and J. A. Levine, "Topology-preserving meshing of implicit scalar fields via monotonicity constraints," in *2026 IEEE Visualization and Visual Analytics (VIS)*, 2026.
- 3 J. Li, T. A. J. Ouermi, M. Han, and C. R. Johnson, "Webuvisbox: A scenario-based framework for collaborative and accessible 3d uncertainty visualization on the web," *Computer Graphics and Applications*, 2026.

- 4 Q. Wang, J. Li, P. J. Moos, and A. Venosa, "A single-cell atlas and shiny-based framework for murine lung injury and remodeling," *Frontiers in Bioinformatics, section Single Cell Bioinformatics*, 2026.
- 5 X. Yan, J. Li, and B. Wang, "Locomapper: Local structural uncertainty analysis of mapper graphs under cover perturbations," in *Uncertainty Visualization Workshop 2026*, 2026.

Published Journal Articles

- 1 J. Hart, B. van Bloemen Waanders, J. Li, T. A. J. Ouermi, and C. R. Johnson, "Hyperdifferential sensitivity analysis with respect to model discrepancy: Prior distributions," *International Journal for Uncertainty Quantification*, vol. 16, no. 1, pp. 51–77, 2026, ISSN: 2152-5080. [DOI](#): 10.1615/Int.J.UncertaintyQuantification.2025059589.
- 2 M. Berger, J. Li, and J. A. Levine, "A generative model for volume rendering," *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 4, pp. 1636–1650, 2019. [DOI](#): 10.1109/TVCG.2018.2816059.

Conference Proceedings

- 1 J. Li, T. A. J. Ouermi, M. Han, and C. R. Johnson, "Uncertainty tube visualization of particle trajectories," in *2025 IEEE Workshop on Uncertainty Visualization: Unraveling Relationships of Uncertainty, AI, and Decision-Making*, 2025, pp. 1–11. [DOI](#): 10.1109/UncertaintyVisualization68947.2025.00005.
- 2 M. Han, T. M. Athawale, J. Li, and C. R. Johnson, "Accelerated depth computation for surface boxplots with deep learning," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 38–42. [DOI](#): 10.1109/UncertaintyVisualization63963.2024.00009.
- 3 M. Han, J. Li, S. Sane, S. Gupta, B. Wang, S. Petruzza, and C. R. Johnson, "Interactive visualization of time-varying flow fields using particle tracing neural networks," in *2024 IEEE 17th Pacific Visualization Conference (PacificVis)*, 2024, pp. 52–61. [DOI](#): 10.1109/PacificVis60374.2024.00015.
- 4 S. Jeong, J. Li, C. R. Johnson, S. Liu, and M. Berger, "Text-based transfer function design for semantic volume rendering," in *2024 IEEE Visualization and Visual Analytics (VIS)*, 2024, pp. 196–200. [DOI](#): 10.1109/VIS55277.2024.00047.
- 5 J. Li, T. A. J. Ouermi, and C. R. Johnson, "Visualizing uncertainties in ensemble wildfire forecast simulations," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, IEEE, 2024, pp. 84–88.
- 6 T. A. J. Ouermi, J. Li, T. M. Athawale, and C. R. Johnson, "Estimation and visualization of isosurface uncertainty from linear and high-order interpolation methods," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 51–61. [DOI](#): 10.1109/UncertaintyVisualization63963.2024.00012.
- 7 T. A. J. Ouermi, J. Li, Z. Morrow, B. Van Bloemen Waanders, and C. R. Johnson, "Glyph-based uncertainty visualization and analysis of time-varying vector fields," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 73–77. [DOI](#): 10.1109/UncertaintyVisualization63963.2024.00014.

- 8 J. Li, D. Van Boxel, and J. A. Levine, "Autoencoder-aided visualization of collections of morse complexes," in *2022 Topological Data Analysis and Visualization (TopoInVis)*, 2022, pp. 18–28. [DOI: 10.1109/TopoInVis57755.2022.00009](#).
- 9 Z. Wang, D. Cashman, M. Li, J. Li, M. Berger, J. A. Levine, R. Chang, and C. Scheidegger, "Neuralcubes: Deep representations for visual data exploration," in *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 550–561. [DOI: 10.1109/BigData52589.2021.9671390](#).

Thesis

- 1 J. Li, "Deep learning approaches for exploring collections of visual features of scalar fields," Ph.D. dissertation, The University of Arizona, 2022.

Presentations & Posters

Public Presentations

- 1 J. Li, *From coin flips to deep ensembles: A practical guide to machine learning uncertainty*, Invited talk at the AIX@TACC, University of Texas at Austin, 2026.
- 2 J. Li, T. A. J. Ouermi, and C. R. Johnson, *Visualizing uncertainties in ensemble wildfire forecast simulations*, Oral presentation at the IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks, 2024.
- 3 J. Li, *Visualizing collection of data*, Invited talk at the University of Utah, 2023.
- 4 J. Li, *Deep learning approaches for exploring collections of visual features of scalar fields*, Invited talk at the SCI Institute, University of Utah, 2022.
- 5 J. Li, D. Van Boxel, and J. A. Levine, *Autoencoder-aided visualization of collections of morse complex*, Oral presentation at the IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis), 2022.

Posters

- 1 J. Hart, B. van Bloemen Waanders, J. Li, T. A. J. Ouermi, and C. Johnson, *ASCR AIVIS: Visualization insight for hyper-differential sensitivity analysis*, Poster presented at SCI X (Also presented in ASCR PI meeting in Texas), 2025.
- 2 M. Han, J. Li, S. Sane, S. Gupta, B. Wang, S. Petruzza, and C. R. Johnson, *Interactive visualization of time-varying flow fields using particle tracing neural networks*, Poster presented at SCI30, 2024.
- 3 J. Li, T. A. J. Ouermi, and C. R. Johnson, *Visualizing uncertainties in ensemble wildfire simulations*, Poster presented at SCI30, 2024.
- 4 T. A. J. Ouermi, J. Li, T. M. Athawale, and C. R. Johnson, *Approximation and visualization of surface uncertainty*, Poster presented at SCI30, 2024.
- 5 J. Li, P.-T. Bremer, M. Larsen, C. Harrison, and J. A. Levine, *Integration between BabelFlow and ascent for in-situ data analysis*, Poster presented at the LLNL student poster symposium, 2019.